

Competitive Uno!

Hopefully you're all familiar with the common card game Uno. You begin with 7 cards in your hand, and the aim of the game is to play all your cards before your opponents do. You can find a full set of the rules at <https://www.unorules.com/>. We will be playing the 108 card version with 25 cards of each colour, 4 Wild and 4 Wild Draw cards.

Your task for this Workshop is to implement a strategy for your Python Uno Player (PUP) to play cards! Work together as a team and create the best strategy you can. Towards the end of the Workshop, we will **have your PUP's face off against each other and see who has the best PUP!**

We have simplified this *as much as possible* to make it easy to do within the workshop. You have been given the skeleton of a `UnoPlayer` class, with most methods already implemented. You only need to:

- Change the `TeamName.py` file to your team name, you will make your changes in this file.
- Assign an instance variable called `name` with the name of your team in the constructor (not through a parameter).
- Implement the `play_card()` method to play a card from your hand, represented by the `hand` instance variable.
- **You do not need to** worry about adding or removing cards to/from your hand. This is all controlled by the simulation. You **only choose a card to play**. Assume your hand begins with cards in it, and any required draw is completed for you.

Cards will be represented by strings. For example:

- A blue 9 card will be: `"blue 9"`
- A yellow draw card will be: `"yellow draw"`
- The only cards without colour will be:
 - The Wild card: `"wild"`
 - The Wild Draw card: `"wild_draw"`
 - However, when you play them, you will add a colour (see below)

Your `play_card` method should have **two parameters**:

- A **string** that represents the current top card of the discard pile.
 - It will always be in the format `'colour type'`
 - Even for Wild or Wild Draw (see below)
- An **integer** that represents the number of cards you are expected to draw. If the previous player(s) have not played Draw 2 or Wild Draw (or they have picked it up) this will be **zero**.
- You should return a **string** that represents the card you would like to play or `None` if you would not like to (or can't) play a card.
- You must make sure you make a **valid play**:
 - The card must be one from your hand.
 - The card must be able to be played on the current card on the discard pile.
 - If it is a Wild or Wild Draw card, you must add a colour to it before returning it from the method. For example, to play a Wild card and change the colour to blue, you would return `"blue wild"`.
 - If you select an invalid card, you will instead draw two cards and your turn will end.
 - Returning `None` is a valid play, and would result in a single pick-up instead of penalty.

There will be two types of winners determined: Those who **win the most rounds of Uno** (i.e. running out of cards first) and those **who have the least total points** at the end of the simulation. These might not always be the same team, depending on the strategy you use!

Our rules:

We will attempt to follow the rules for the 108 card version of Uno. There are some caveats to make it easier for you to implement:

- The card that begins the game at the top of the discard pile **will not** be a Wild or Wild Draw card.
- If the card that begins the game at the top of the discard pile is a Draw card, you will not have to draw.
- You do not have to implement “Yelling Uno” when you have one card remaining.
- You can assume there will be at least 4 players in a game, and no teams.
- You can play any “Draw” card to continue a draw stack and not have to draw yourself. For example, you can play a Wild Draw on a Draw 2.

Points system:

- We will simulate 1,000 rounds of Uno.
- You want to have the lowest points at the end of the simulation. Obviously, the winner of a round receives 0 points for 0 cards. The remaining players will have their points summed and added to their total over the simulation.
 - Numbered cards: Worth their face value 0-9
 - Draw Two/Skip/Reverse: Worth 20 points
 - Wild/Wild Draw: Worth 50 points each.